ONE-PAGE BRIEF

# Post-quantum cryptography review: frost @ 0966bd1529aa

Client: Zcash Foundation — public repository (passive review) · Source: github.com/ZcashFoundation/frost, commit `0966bd1529aa` · Reviewed: 2026-10-30 22:48 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
**Level 0 of 4**
no post-quantum cryptography

FINDINGS
**0** Critical **0** High **5** Medium
**3** Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | |
|---|---|
| Medium<br>QSI-2026-08318 | **No hybrid post-quantum key exchange configured (book/src/zcash/server.md)**<br>What to do: With OpenSSL ≥ 3.5 (or BoringSSL/rustls with ML-KEM), add X25519MLKEM768 ahead of X25519 in the groups list after interoperability testing. |
| Medium<br>QSI-2026-08319 | **Quantum-vulnerable FROST: threshold signing and MPC protocols (27 file(s))** preliminary<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium<br>QSI-2026-08320 | **Quantum-vulnerable RedDSA: threshold signing and MPC protocols (6 file(s))** preliminary<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium<br>QSI-2026-08321 | **Quantum-vulnerable ECDSA: threshold signing and MPC protocols (2 file(s))** preliminary<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |

4 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium threshold signing and MPC protocols | FROST, RedDSA, ECDSA | Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the... |
| Medium role not determined | ECDSA | Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)... |

## KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 4; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 3.

Static review of source code and manifests; files analysed: 61. No requests were sent to the owner's systems.

Full report, content SHA-256: `58a7a08c895960a650138e494c874d062a1f0407e7503763db5e25263c72a421`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.